

Xiuer Gu

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xiuerg.github.io/xiu-portfolio

Research Interests

Human-Computer Interaction; Human-Centered AI; Generative AI for Interactive and Creative Systems; Digital Health and Accessible Technologies

Education

The University of Texas at Austin, Austin, TX, USA

Aug 2024 – May 2026

M.S. in Information Studies; GPA: 3.87/4.0

Selected Coursework: Human-Computer Interaction; AI in Health; Computer Vision; Fine-Tuning Open-Source LLMs; Natural Language Processing / Applications; Human Computation & Crowdsourcing

The Ohio State University, Columbus, OH, USA

Aug 2021 – May 2024

B.F.A. in Art & Technology; GPA: 3.7/4.0

- Dean's List: Spring 2022, Spring 2023, Autumn 2023

Publications

1. C. Chow[†], X. Gu[†]. "Same Prompt, Different Care? Evaluating Inequities in AI-Generated Mental Health Support for Immigrant Communities." *Journal of the Society for Social Work and Research*, 2026. [[Paper Link](#)]
2. J. Wu, X. Gu, Z. Li, Y. Jin, Y. Diao, Z. Li, Z. Song, X. Zhang, and Z. Fan. "HieroAction: Hierarchically Guided VLM for Fine-Grained Action Analysis." Preprint, 2025. [[Paper Link](#)]

[†]Equal contribution.

Selected Projects

Human-Centered Digital Support for Written Exposure Therapy

Spring 2026

Capstone Project, UT Austin; Supervised by Dr. Ying Ding

- Translated an AI-based Written Exposure Therapy simulation and evaluation pipeline into an interactive research application for configuring, observing, and reviewing simulated therapist-patient sessions.
- Designed safety-aware interactions for trauma-related symptoms, addressing AI boundaries, distress escalation, pacing, privacy, and cognitive load.
- Led interaction design and front-end development, turning therapeutic and research requirements into a reusable prototype for future evaluation.

CoReLink: AI Resource Navigation for Dementia Caregivers

Spring 2025

UT Austin School of Nursing, NURSING-AI Challenge; Supervised by Dr. Bo Xie

- Deployed a local LLM and integrated external search APIs to extract care-related needs from caregiver narratives and clinical notes and support downstream resource retrieval.
- Developed a multi-stage pipeline for need extraction and location-based retrieval of medical and community resources tailored to care needs and access constraints.
- Prototyped a clinician-facing interface for reviewing resource recommendations, refining identified needs, and planning follow-up actions.

Automated Thematic Analysis of AAOCA Interviews with Multi-Agent LLMs

Spring 2025

Research Project, UT Austin; Supervised by Dr. Ying Ding

- Developed a multi-agent LLM pipeline for end-to-end inductive analysis of parent interviews, transforming source evidence into codes, subthemes, and themes.
- Evaluated the pipeline across four temperature settings and designed "Parallel Critic Meetings" to compare, cluster, and consolidate independently generated analyses.

International Projection of China's National Geographical Indication Products in Hunan Province

Fall 2024

Hunan University of Technology & Business

- Collaborated with Prof. Ling Zhang and the Hunan Intellectual Property Bureau on a funded research project (Approval Code: 2024F007K) investigating strategies for the international promotion of Hunan's geographical indication products.

Research Experience

Moritz Center for Societal Impact UX Research / Interactive Experience Designer

Feb 2026 – May 2026

Part Time, Austin, TX, USA

- Conducted qualitative stakeholder interviews to identify audience needs and shape the exhibit's content strategy, narrative structure, and interactive storytelling approach.
- Translated research insights into interaction flows and front-end experiences, leading the design and development of a project website and four interactive storybooks using Figma, React, and Next.js.
- Iteratively refined information architecture and interface usability through stakeholder feedback and supported deployment at a public exhibit attended by more than 200 visitors.

The University of Texas at Austin Graduate Researcher

Jul 2025 – May 2026

Graduate Portfolio in Ethical AI

- Selected for UT Austin's Graduate Portfolio in Ethical AI and developed a research project on human value alignment in autonomous systems.
- Designed scenario-based studies examining how people prioritize safety, accessibility, privacy, fairness, and efficiency in campus delivery robot decisions.
- Developed a human-in-the-loop preference-learning and transparency framework to translate contextual judgments into interpretable system behavior.

The Ohio State University Research Assistant

Sep 2023 – May 2024

Living Art Eco Lab, Columbus, OH, USA

- Created 3D models of lichen structures for ecological visualization and the "In a Hotter House" exhibition.
- Developed a TouchDesigner installation integrating Arduino-based thermal imaging sensors to simulate climate impacts for more than 100 visitors.

Hunan University of Technology & Business Research Assistant

Jun 2023 – Aug 2023

Changsha, China

- Conducted field interviews with local artisans and translated findings into bilingual storyboards and digital narratives.
- Supported cross-cultural adaptation and post-production for a UNESCO-sponsored heritage video archived by the UN, balancing accessibility and cultural authenticity.

Skills

- **Programming & Development:** Python, JavaScript, React/Next.js, HTML/CSS
- **AI & Computational Methods:** LLM Fine-Tuning, RAG, Multi-Agent LLM Workflows
- **HCI Research Methods:** User research, qualitative interviews, usability evaluation, accessibility evaluation, mixed-methods research
- **Design & Prototyping:** Figma, Unity, Blender, TouchDesigner
- **Languages:** Chinese (native), English (fluent)